On the Design and Validation of the MISSat-1 Small Satellite Ground Station

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ABSTRACT

Necessary to proper communications with any small satellite is a working and capable ground station. The design of these stations constitute proper selection of multiple essential components including; antennas, preamplifiers, rotators, cables, and transceivers. The combination of these elements ensure that the power received by the small satellite, and likewise the power received from the satellite, is sufficient for proper reception. This paper will examine both the minimum necessary components and ones selected for the MISSat-1 ground station using the Friis transmission equation and other principles of telecommunications. In addition, the ground station intended to support the communication needs of the Mississippi Imaging Space Satellite (MISSat-1) are be measured and validated with common equipment. The 2MCP14 and 436CP30 source antennas of the ground station are evaluated by use of a spectrum analyzer, rotator, and computer software to determine the far-field pattern. With this data, the performance of the ground station is determined. The measurements generated are compared with manufacturer specifications of common ground station antennas. Although this ground station is within the operating ranges of small satellite communication, the minimum requirements of such a station are analyzed. By investing more in the performance of a ground station several advantages can be obtained such as transmitting with less loss or with a longer window of communication.

1. Overview of Ground Station Components

The goal for any operational small satellite ground station is to be able to effectively transmit to and receive from orbiting satellites. In order to ensure proper communications, the selection of necessary components becomes of utmost importance. This paper will explore the construction and validation of a ground station with both 2m (146 MHz) and 70cm (436 MHz) capabilities which are the most common uplink and downlink bands for small satellites.

The first component(s) in the ground station to be considered are the antennas for the chosen bands. Like many of the other components, there are a wide variety of choices for purchase or building. Most small satellite ground stations use a multiple element Yagi-Uda as their antenna for the 2m and 70cm bands. These antennas can provide gain anywhere from 6 to 24 dB over an isotropic radiator depending upon their configuration and number of elements. While larger gain can be advantageous in receiving weaker signals and transmitting more efficiently, increasing gain will decrease the beam width of the transmitted or received signal. As this beam width decreases, the antennas become increasingly sensitive to the accuracy of aiming or rotator system, thereby making it harder to aim. Selection of a proper antenna must also consider the accuracy of the system to aim the antennas in addition of the gain desired from the antenna.

The rotator or azimuth/elevation control system has a very simple purpose, to aim the antenna system effectively at the orbiting satellite and to keep it aligned as the orbit passes through its field of view. Any appropriate rotator system should be able to aim the antennas anywhere in the half-hemispherical area that a satellite may orbit through. While many different rotators are available, they are mostly differentiated by both cost and aiming accuracy. Rotators may be acquired with very fine accuracy but also with high cost, while cheaper rotators may be easily found but suffer with aiming accuracy. This balance must be weighed with the beam width of the selected antennas to ensure constant and effective communications.

Preamplifiers may be included in any ground station to increase the ability to receive weak signals from satellites that may be transmitting in low power or may be near the horizon. While many satellites are easily received without the need for preamplifiers, they can aide in extending the time/range that satellites may be communicated with. Preamplifiers may be easily and cheaply procured for both the 2m and 70cm bands with little trouble and offer up to 20dB gain with very low noise if correctly installed adjacent to the antennas.

Selecting a proper cable or transmission line system is easily just as important as any other component in the ground station as it becomes the major source of controllable loss. Care must be taken in selecting proper coax to minimize these losses. Many different coax types are available with a range of costs extending from the very cheap but lossy to the very expensive and very low loss. This loss is directly a function of the coax type, length of cable needed, and the frequency used. As a general rule for any type of coax, the losses will increase as the frequency increases. Selection of proper coax can often help mitigate signal loss situations.

The last link in the ground station chain is often the transceiver or communications receiver/transmitter. For most small satellite ground stations this is often a commercially made amateur radio transceiver. While only a handful of models are currently available they generally offer comparable performance in both transmission and receive specifications for 2m and 70cm bands. The major difference between the various models is generally only their price.

1.1. Cost Comparison of Basic Components

Each university seeks to optimize the cost vs. the functional requirements. Some instrumentation has addition features which add to the cost. For example, the University of Texas chose a more expense dish-type antenna [2]. Their antenna performs in the UHF/VHF/S bands. The larger more capable dish antenna required profession installation and engineering consultation. This alone cost $24,000.00. Decisions like this help determine the final cost of a ground station.

Table 1: Sample cost comparison

	Antenna	Transceiver	Preamplifiers	Coax & connectors	Rotator
University of Arizona[3]	720.00	1,450.00	1203.00	374.95	260.00
National Cheng Kung University[4]	575.00	3030.00	263.00	700.00	1,060.00
University of Hawaii[5]	358.00	870.00	330.00	20.00	639.95
University of Texas[2]	20,000.00	1,500.00	275.00	4,500.00	
Clyde Space[6]	1,033.8				1,239.95

2. University of Mississippi Ground Station Components

There are many choices for each component of the ground station. Different components were compared based on cost and functionality. For example, antennas may be chosen that provide more directivity than others but at the sacrifice of a smaller beam width. Transceivers with more sensitive receive capabilities be chosen but at higher cost. However, the cheaper cost of a preamplifier may be of more importance overall. Each of the possible situations may be explored by utilizing the Friis transmission equation [1].

$$\mathrm{P_r} = P_t + G_t + G_r - L_t + 20log_{10}(\frac{\lambda}{4\pi R}); \text{ units in decibels} \qquad (1)$$

The goal is to maximize the power received (P_r) given the transmission constraints. In the equation, P_t represents the effective power transmitted, $G_t$and $G_r$are the gain of the transmitting and receiving antennas respectively, L_t represents any additional loss or gain that may be in the system (coax loss, polarization loss, preamplifier gain, etc), and the last term is the path loss by sending the signal through a distance of R. Although this type of equation might be solved as a linear program, the discrete functionality of the instruments and limited choices due to budget limitations allow manual selection from an affordable set of components.

2.1. Antennas

Antennas with a broad beam width were chosen to allow less accuracy while tracking satellites but still fulfill the primary mission. The 2MCP14 and 436CP30 circularly polarized Yagi-Uda antennas from M2 Antennas, were selected based on price and specific performance parameters, such as operating frequency, half power beam width, gain, and size. Circularly polarized antennas were chosen due the ability to be unaffected by polarization angle. This fact far outweighs the 3dB of loss due to receiving linearly polarized signals.

The VHF antenna operates with a center frequency of approximately 145 MHz in the amateur radio band. The optimum band width is between 143MHz and 148MHz with a gain around 12.34 dBic and beam width of 52 degrees circular. The front to back ratio is 20dB. [7]

The UHF antenna operates with a center frequency of approximately 436 MHz in the amateur radio band. The optimum band width is between 432MHz and 440MHz with a gain of 15.50 dBic and beam width of 30 degrees circular. The front to back ratio is 18dB. [8]

Figure 1: 436CP22 70cm, cicularly polarized, 22 element Yagi-Uda antenna (Left).
2MCP14 2 meter, circularly polarized, 14 element Yagi-Uda antenna (Right).

2.2. Preamplifiers

The SP-2000 (2m) and SP-7000 (70cm) low noise, mast mounted, gallium-arsenide FET preamplifiers from SSB Electronic USA are used to amplify the power of communication signals. These amplifiers provide a signal gain between 10-20 dB within their respective frequency band. [9]

Figure 2: SP-2000 / 7000 preamplifiers mounted on the mast.

2.3. Rotator

The Yaesu G-5500 was chosen as the azimuth/elevation rotator. An H-frame was used to mount the 2 meter and 70 cm antennas to the rotator. The G-5500 controller box is controlled using an Arduino Duemilanove. A computer using Ham Radio Deluxe satellite tracking software is used to send commands to the Arduino, controlling the pointing of the antennas.[10]

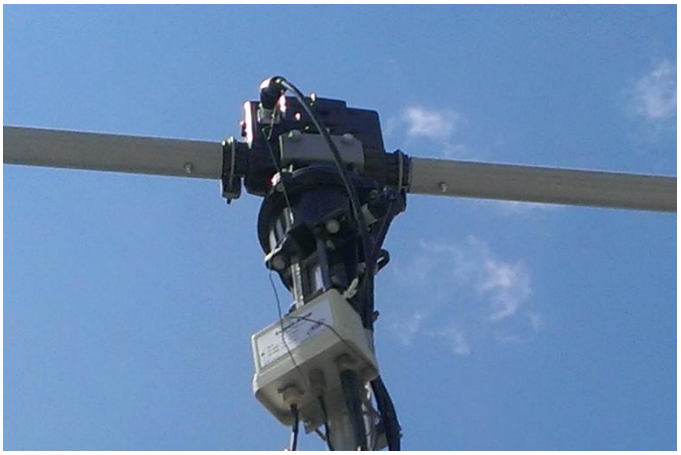


Figure 3: Yaesu G-5500 Azimuth/Elevation rotator mounted on top of the mast.

2.4. Cables

CNT-600 is used for transmission between the transceiver and the mast mounted preamps. CNT-240 is used to connect the preamps and antennas. The CNT-600 attenuation in decibels per 100 feet is 0.97dB and 1.71 dB at 150MHz and 450MHz, respectively. The CNT-240 attenuation in decibels per 100 feet is 3.0 dB and 5.3 dB at 150MHz and 450MHz,

respectively. Runs of 100ft were used for both bands from the preamplifier to the transceiver, while runs of 10ft were used from the antennas to the preamplifiers. The combined loss of the coax for the system is 1.27 dB at 150 MHz, and 2.24 dB at 450 MHz. [11]

2.5. Transceiver

The Kenwood TS-2000 is the transceiver chosen for communication with small satellites. The TS-2000 is a HF/VHF/UHF transceiver with a 100 watt maximum output power. This transceiver provides adequate power while remaining relatively low cost. [12]

Figure 4: Kenwood TS-2000 VHF/UHF transceiver.

2.6. Component Costs

As can be seen from Table 2, the ground station cost was kept below $6000. This cost only includes the minimum essential components needed to communicate. It could be argued that the computer to process data and control tracking should be included, but this cost was not included in Table 1 for comparison.

Table 2: Itemized cost of ground station components

ITEM	MODEL	USD
UHF Antenna with R/L Switch	2MCP22	$443.00
VHF Antenna with R/L Switch	436CP30	$468.00
Preamplifiers	SP-2000/7000	$330.00/$330.00
Rotator	Yaesu G-5500	$699.95
Roof-Top tower	RT-832	$622.00
Coax	CNT-600/240	$356.90/$43.20
Radio	Kenwood TS-2000	$1474.99
Radio PSU	Kenwood KPS-15	$139.99
Misc Parts	Cables, antenna supplies, etc	$746.38
Total Ground Station Cost		**$5654.41**

3. University of Mississippi Ground Station Performance

This section will present the performance of the ground station and the steps taken to perform a measurement using common equipment.

3.1 Measurement Preparation

The necessary hardware was assembled on the roof of the engineering building but only limited capability control software was available. It was decided to write new software to control hardware and instrumentation while collecting data autonomously.

3.1.1. Program

The program for rotator and spectrum analyzer control was written using MATLAB. The program begins by opening communication ports with a spectrum analyzer and the rotator controller, an Arduino Duemilanove. Then initial parameters are set, such as frequency and starting azimuth angle. The rotator is then commanded to rotate to 360 degrees, during which time the position of the rotator and amplitude of a received signal is read multiple times. Several data sets are collected for a single frequency to have many data points to analyze. The data sets are then averaged together and plotted using MATLAB.

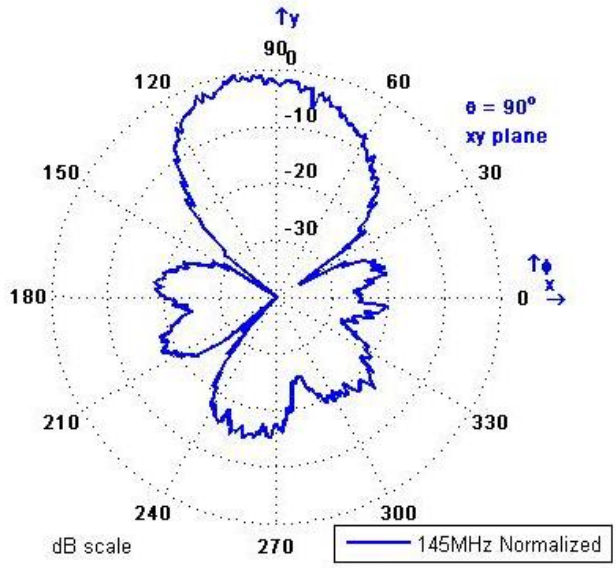
3.1.2. Antenna Under Test and Source Antenna Setup

A small VHF/UHF antenna in the far field zone was connected to a signal generator set to a frequency within the band of the ground station antenna being measured. One of the receiving antennas, such as the VHF antenna, was connected to the spectrum analyzer to be measured using the new software. The receiving antenna under test was changed, the appropriate frequency was chosen on the signal generator, and more measurements were taken.
(insert figure here)

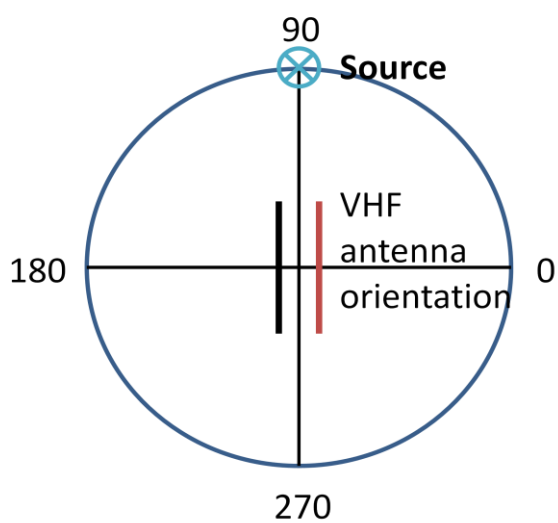
3.2. Antenna Performance

3.2.1. VHF Antenna

The VHF antenna was measured at a frequency of 145MHz, which is about the center frequency of the optimum band width for the antenna. The radiation pattern measurement was performed in the orientation shown on the VHF antenna represented by the red bar and the source antenna represented as the blue ‘X’ in Figure 5b.



a) Normalized radiation pattern of VHF antenna at 145MHz



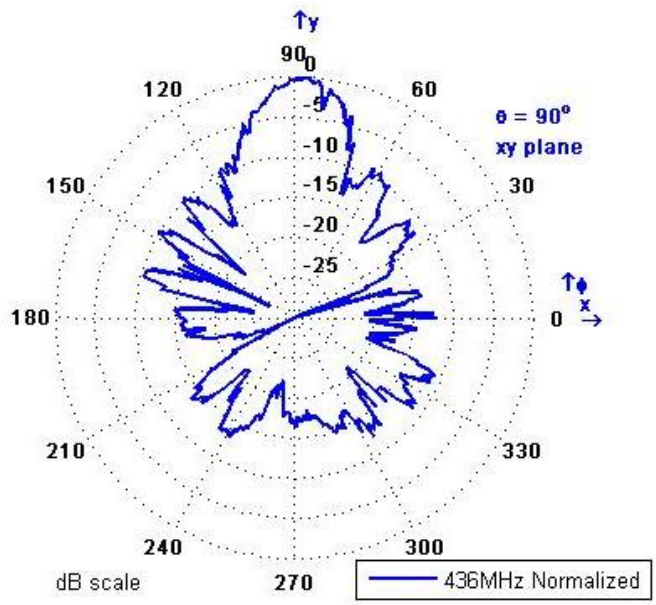
b) VHF antenna orientation in relation to source (x) antenna

Figure 5: VHF antenna (a) measured radiation pattern and (b) orientation.

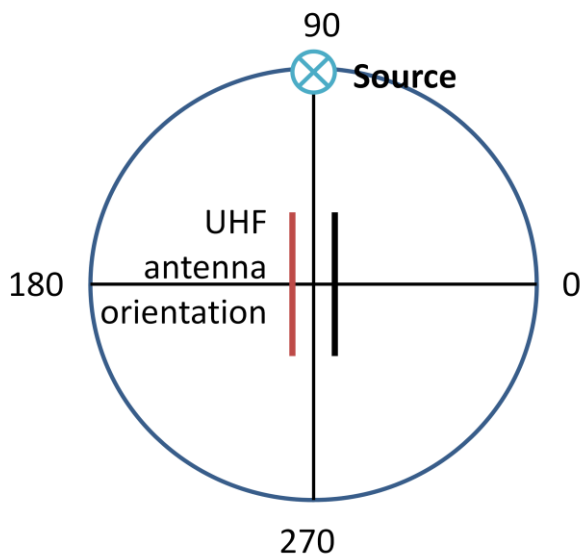
Figure 5a shows a half power beam width of about 33 degrees. The minor lobes occur below -15dB of the peak. The front to back ratio is about 15db.

3.2.2. 436CP30

The 70 cm antenna was measured at a frequency of 436MHz, about center frequency of the antenna's optimum band width, with preamplifiers turned off.



a) Normalized radiation pattern of UHF antenna at 436MHz



b) UHF antenna orientation in relation to source (x) antenna

Figure 6: UHF antenna (a) measured radiation pattern and (b) orientation.

Figure 6 shows a half power beam width of about 22 degrees. The side lobes occur below -9dB of the peak. The front to back ratio is ~12 db.

3.2.3. Comparison with Manufacturer Specifications

The manufacturer specifications suggest the normalize radiation pattern should resemble the pattern in Figure 7. Much of the difference between the radiation patterns in Figures 6a and 7 are due to different scaling and the offset of the source antennas height below the receiving antenna's xy-plane.

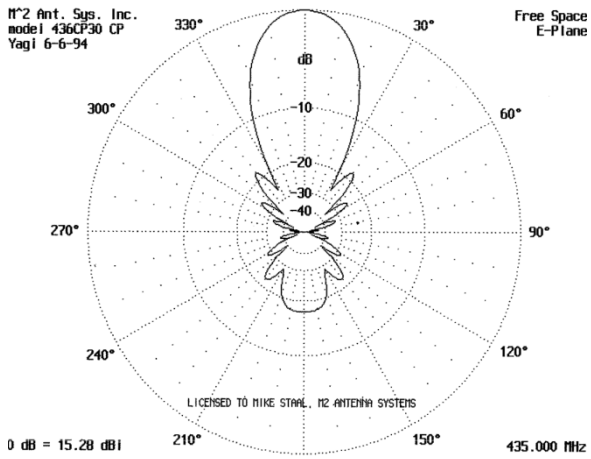


Figure 8: Normalized radiation pattern of UHF antenna at 435 MHz [13]

4. Contact Established with Small Satellites

We were able to make contact with two low earth orbit amateur radio FM repeater satellites operating in the UHF/VHF bands. The first one was Amsat-Oscar 7 (AO-7) and the other was SO-50. We were able to have each satellite repeat our VHF transmission to other amateur radio enthusiast. The reception of the repeated transmission was verified over UHF from the satellites.

5. Conclusion

This paper has shown that using minimum resources, it is possible to build a low cost ground station capable of transmitting and receiving to small satellites in low earth orbit. With this capacity, future research satellite projects can be accomplished. One such project is underway at the University of Mississippi where students are building a CubeSat class satellite to capture terrestrial images.

Acknowledgments

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